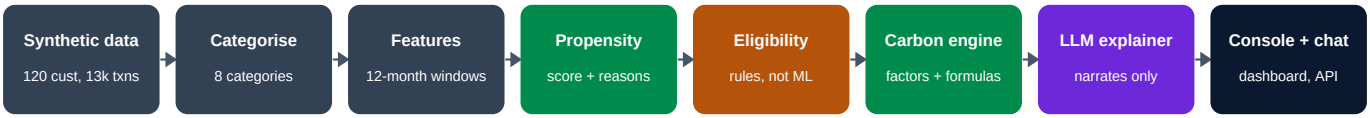
Team 292

Final Report: Green Finance Assistant

Megathon X: Singularity | PS1 Green Customer Insights Platform. An explainable green-finance intelligence platform for retail banking in India. All data synthetic; all impact figures are estimates.

1. Solution and architecture

Green Finance Assistant turns raw bank transactions into explainable green-finance opportunities for the four required products (EV Finance, Green Home Loan, Solar Finance, Paperless Banking) plus the two optional ones (Green Investments, Green Personal Loan). The design rule from the brief is enforced in code, not just in prose: **the ML layer predicts, a deterministic rules engine decides eligibility, a rules-based carbon engine calculates impact, and the LLM only narrates structured outputs.** Nothing downstream can alter anything upstream.



ML predicts | rules decide eligibility | carbon engine calculates | LLM only explains | memory only re-orders

Requirement	How it is met	Evidence
FR1 Customer 360	One record per customer joining profile, vehicle, energy, digital-usage, products held and 12-month transaction features.	Console > Customers; /api/customers/{id}
FR2 Categorisation	Exactly one of 8 mutually exclusive categories per transaction; documented keyword rules for uploaded CSVs; duplicates flagged and excluded from features.	100% coverage; 242 duplicates flagged
FR3 Opportunities	Per customer x product: propensity, confidence, reason codes with evidence strings, data timestamp.	720 pairs scored, 556 eligible, 79 recommended
FR4 Impact	Deterministic carbon engine with a factor registry (CEA grid, IPCC, DEFRA); formula, assumptions and inputs are returned with every figure; reported as annual kg CO_2e estimates.	Carbon Engine page; /api/carbon/calculate
FR5 Customer insight	LLM narration from the structured payload only; deterministic template fallback if no key or on any failure; grounded chat companion.	Companion; /api/explain
FR6 Business insights	Console: KPIs, opportunities by product, segment table, propensity distribution, monthly spend trends, estimated CO2e and indicative value, drill-down Customer > Opportunity > Evidence > Impact > Action.	Console Overview, Opportunities
FR7 Explainability	Each recommendation shows reason codes, evidence, confidence, timestamp; ineligible products are stored for audit but never recommended.	0 violations across 720 pairs

Portfolio view (synthetic): 79 recommended opportunities with an estimated 78.7 t CO_2e per year gross avoided if all were adopted (an upper bound, not a forecast; adoption probability and rebound are not applied).

2. Data, labels and validation design

- **Data (synthetic, seed 42):** 120 fictional Indian customers, 13,110 transactions over 12 months with raw merchant descriptions, six hand-built featured personas for demos, injected duplicates for the data-quality path. No real or confidential banking data. Generation is deterministic (regeneration checksum is part of the test suite).
- **Take-up labels:** a documented rule with different weights from the scorer: sigmoid of weighted signals + N(0, 0.5) noise, seeded per customer-product pair, and **conditioned on eligibility** (an ineligible customer cannot take up a product, so those rows are 0). Take-up is one synthetic flag per pair; the outcome window (assumed 12 months after the observation period) is not separately modelled.
- **Splits:** customer-level, disjoint, by sha256(customer_id) mod 10: train 60% / validation 20% / test 20% (432 / 120 / 168 customer-product rows). Plus 5-fold customer-level CV over all 120 customers (~28 test rows per product is small).
- **Models compared:** (a) transparent scorer, i.e. documented hand-set weights + sigmoid, whose score decomposes into reason codes; (b) L2 logistic regression (Newton's method) trained on the train split only, same features; (c) strongest single-signal heuristic per product; (d) random (0.5).

3. Results (held-out test split, 95% bootstrap CIs, 1,000 seeded resamples)

Product	n (pos)	Scorer AUC	95% CI	Logistic reg. AUC	Single-signal AUC	Random	P@k	Brier	5-fold CV AUC (scorer)
EV Finance	28 (9)	1.000	[1.00, 1.00]	1.000	0.854	0.500	1.00	0.103	0.943
Solar Rooftop Finance	28 (7)	0.748	[0.56, 0.90]	0.755	0.524	0.500	0.25	0.185	0.783
Green Home Loan	28 (6)	0.871	[0.70, 1.00]	0.879	0.295	0.500	0.50	0.174	0.786
Paperless Banking	28 (9)	0.918	[0.79, 1.00]	0.959	0.927	0.500	0.75	0.186	0.911
Green Investments	28 (7)	0.830	[0.64, 0.96]	0.789	0.561	0.500	0.50	0.200	0.832
Green Personal Loan	28 (9)	0.863	[0.69, 0.98]	0.778	0.708	0.500	0.75	0.253	0.738
Macro average	168	**0.872**	**[0.81, 0.93]**	**0.860**	**0.645**	**0.500**	**0.62**	**0.183**	**0.830 +/- 0.017**

Macro AUC in 5-fold CV: scorer 0.830, logistic regression 0.799, single-signal 0.628. P@k uses k = max(3, 15% of rows).

- **Baseline comparison.** The scorer beats the single-signal baseline on 5 of 6 products and on macro AUC (0.872 vs 0.645; CV 0.830 vs 0.628). **Paperless Banking is a near-tie** (0.918 vs 0.927): paper-statement volume alone almost determines it, so a multi-signal model adds little. Solar is the weakest product (0.748, CI 0.56-0.90). EV Finance reaches 1.000 on the test split, but with only 28 rows and 9 positives this is a small-sample effect (CV: 0.943), not a claim of perfection.
- **Trained vs hand-set.** The trained logistic regression matches the scorer on hold-out (macro 0.860 vs 0.872) but is weaker in CV (0.799), because ~80 eligible training rows per product is small. It is better **calibrated** (Brier 0.131 vs 0.183; log-loss 0.382 vs 0.536).
- **Calibration weakness (reported, not hidden).** The transparent scorer is systematically under-confident: in the populated bins, predicted vs observed take-up was 0.05 vs 0.19, 0.30 vs 0.58 and 0.49 vs 0.88, and no score exceeds 0.6. Its value is *ranking*; its probabilities should not be read as literal take-up rates. The recommendation threshold (propensity >= 0.35) is therefore a ranking cut-off.

4. Sustainability impact (deterministic carbon engine)

Opportunity	Formula (annual, per customer)	Factors and assumptions	Known-answer test
EV Finance	(km / 15 kmpl x 2.31 kg/l) - (km x 0.14 kWh/km x 0.82 kg/kWh)	IPCC 2006 petrol; CEA grid; 15 kmpl ICE, 0.14 kWh/km EV	15,000 km: 588.0 kg (expected 588.0)
Solar Finance	min(monthly kWh, 3 kW x 120 kWh/kW) x 12 x 0.82	3 kW rooftop; 120 kWh/kW/month; CEA grid	300 kWh/m: 2952.0 kg (expected 2952.0)
Green Home Loan	(annual electricity + gas kg) x 15% retrofit saving	CEA grid 0.82; IPCC gas 2.04 kg/SCM; 15% saving	464.8 kg (expected 464.8)
Paperless	(statements x 0.020 kg) + (branch visits x 0.44 kg), annualised	DEFRA 2023 paper; 4 km petrol 2-wheeler visit	11.0 kg (expected 11.0)

*Impact is gross annual avoided CO_2e (project/avoided-emissions style, consistent with GHG Protocol Scope 3 transparency principles; not a PCAF financed-emissions disclosure). Green Investments is deliberately **not quantified** (PCAF portfolio data not available), and Paperless is shown separately as digital adoption, which is not itself a carbon reduction. Rebound effects and behaviour change are not netted off, so true savings will be lower.*

5. GenAI layer, customer memory and testing

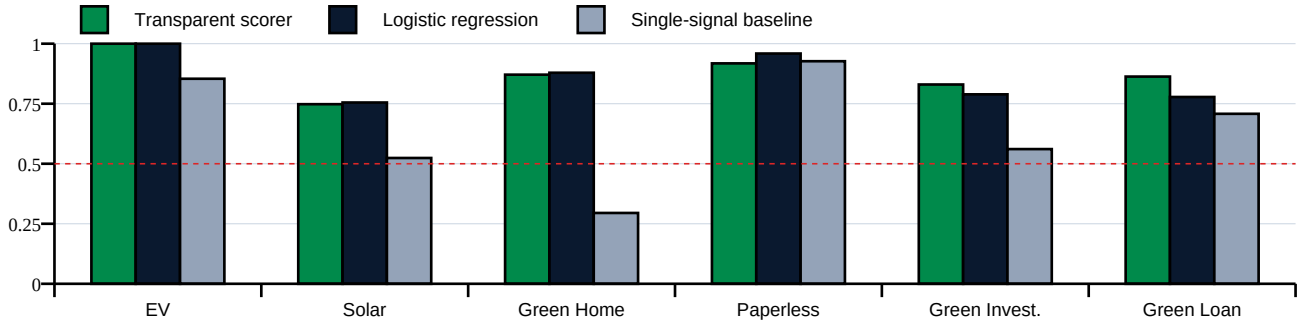
- **LLM explanations.** The prompt receives only the stored structured payload (propensity, confidence, reason codes, estimated impact + formula, data timestamp). The system prompt forbids new numbers, eligibility or approval claims, and requires the word 'estimated', under 110 words, one 'Next step:' action line. The model is called via API (optional); without a key, or on any error, a deterministic template renders the same payload. Explain-endpoint is read-only over stored outputs.
- **Companion chat** answers only from a per-customer facts bundle (top spend categories, held products, recommended opportunities, estimated impact, recent feedback) with a guardrail line that estimates are never guarantees.
- **Customer memory (optional extension).** Responses (viewed / interested / not for me / dismissed) are stored with fixed weights and used *only to re-order* recommendations; each item keeps its original propensity, confidence, eligibility and impact and gains a transparent memory annotation (boosted / suppressed / neutral, rank shift).
- **Automated tests:** 31 pytest tests pass (carbon known-answers; eligibility rules; guardrails; fallback determinism; number-grounding; reproducibility; CV coverage). Live harness: carbon 4/4 known answers, guardrails 3/3 (0 ineligible recommended, 0 recommendations missing impact or evidence), LLM checks 3/3. Two consecutive evaluation runs are identical; full evaluation (incl. 5-fold CV and 1,000-resample bootstraps) runs in ~456 ms.
- **Scenario robustness.** Missing vehicle: EV propensity floored (0.03) and no impact, so never recommended. Existing EV owner or non-home-owner: ruled out by eligibility rules. Already paperless: no impact, so never recommended. No LLM key or LLM outage: template fallback. A judge-facing *Verify* page recomputes any customer chain live, and a *CSV Upload* sandbox runs the full engine on external data without touching the curated dataset (profile fields are derived from documented heuristics there).

6. Responsible AI note

Area	What we did	What remains / limitation
Privacy	Synthetic, fictional customers only; pseudonymous IDs (CUST-xxxx); no real personal or banking data; informed by DPDP Act 2023 purpose-limitation and data-minimisation principles.	Any real deployment needs lawful basis, DPIA, consent for profiling and access controls, none of which are in scope.
Bias and fairness	Eligibility is a separate, auditable rule set. Scores use no name, city or gender. Age is used only in eligibility rules; income band is used in eligibility and in the Green Investments score.	No per-segment (age, income) metrics were computed, and the synthetic cohort cannot support fairness conclusions. Income-linked rules (e.g. Green Investments needs 10L+) may exclude lower-income customers by design.
Explainability	Interpretable scorer; each recommendation lists reason codes with evidence strings, confidence, data timestamp, the formula and assumptions behind its impact figure.	Confidence is a data-sufficiency heuristic (more transactions and months observed = higher), **not** a calibrated or conformal interval. It says how much evidence exists, not how likely take-up is.
Hallucination	Structured-only prompts, number-grounding test on the fallback narration, forbidden-claims system prompt, deterministic fallback.	Live LLM output is not post-checked automatically; the grounding test covers the fallback template. A numeric post-generation checker for live output is future work.

Area	What we did	What remains / limitation
Human oversight	System recommends and explains only. Eligibility and any product decision stay with the institution's authorised process; figures are labelled as estimates and eligibility as assessed separately.	No human-review workflow is built; feedback is limited to the memory events.
Data limitations	Labels, features and scorer share structure by design, so AUC measures agreement with a documented rule, not real-world take-up. 120 customers give wide CIs.	Scorer weights were rebalanced once after a first evaluation run (EV, Paperless, Green Investments), so hold-out figures are mildly optimistic. Spend-to-physical conversions use fixed Indian price assumptions (INR 7.5/kWh, 105/litre petrol).

Figure: hold-out AUC by product



Test split, 28 rows per product; dashed red line = random (0.5). See Section 3 for confidence intervals. Differences under about 0.1 are within noise at this sample size.

7. User experience

- **Customer Green Companion:** personal scorecard (monthly estimated emissions, digital share), ranked opportunities each with evidence, confidence, estimated annual impact and its method, a clear action (e.g. Explore EV options, Go paperless), streamed LLM explanation, grounded chat, and 'interested / not for me' feedback that feeds customer memory.
- **Lloyds Intelligence Console:** customers analysed, opportunities by product, estimated CO_2e and indicative value, propensity distribution, segment table, searchable customer list and drill-down Customer > Opportunity > Evidence > Impact > Action, down to a single categorised transaction. Emissions, digital adoption and value are reported as separate figures.
- **Trust pages:** Carbon Engine (interactive calculator on the same code path with the factor registry), Evaluation (live metrics), Verify (live recomputation and reproducibility check), CSV Upload sandbox, Methodology and AI-disclosure pages.

8. Limitations and next steps

- **Evidence is synthetic.** Real take-up history is the single biggest gap; until labels come from actual conversions, reported AUCs show ranking agreement with a documented rule only. Next: validate on a public or partner dataset with product-opening labels.
- **Categorisation.** On the synthetic data, categories follow the generator's merchant lists, so 100% coverage is by construction. The keyword rules used for uploaded CSVs have no held-out accuracy yet. Next: a labelled text test set and an ML fallback for the Other class.
- **Probability quality.** Recalibrate the scorer (Platt or isotonic on the validation split) and replace the data-sufficiency confidence with a calibrated or conformal interval.
- **Impact.** Fixed price and efficiency assumptions with no ranges; add low/high bounds, state-wise grid factors, the latest CEA database version, and a PCAF-aligned method for Green Investments.
- **LLM checks.** Add an automated numeric-claim checker on live LLM output (today it is covered for the template fallback), and per-segment fairness reporting.

9. Reproducibility and run instructions

Fixed seed 42; customer-level hashing for splits and folds; seeded bootstrap (123 / 321). Setup, MongoDB and run steps are in README.md. `pytest` needs no database, network or LLM key. `python scripts/make_final_pdf.py` regenerates this report with the live numbers above. The CEA grid factor of 0.82 kg/kWh is a rounded weighted-average baseline; update the factor registry in `backend/carbon.py` to the latest CEA database version before any real use.